

OTT Subscriber Churn Analytics

End-to-end data pipeline & churn intelligence framework integrating subscriber acquisition, contract structures, and support escalation signals to mitigate revenue loss.

Churn & Revenue Impact

Engineered an end-to-end churn analytics pipeline across 20+ KPIs. Uncovered key churn disparities between monthly (55.6%) and annual (8.3%) contract tiers.

Risk Scoring & Segmentation

Built a multi-dimensional risk scoring model incorporating tenure, plan type, and support signals to isolate high-risk cohorts and protect long-term LTV.

Support Intelligence

Joined complaint, CSAT, and escalation data with subscription records to identify root causes like price sensitivity, competitor switching, and service issues.

Calculated Metrics & KPI Engine

KPI METRIC	CALCULATION & LOGIC FORMULA
Churn Rate	<code>churned_customers / total_customers</code>
Churn by Plan / State	<code>churn_rate GROUP BY plan_type country, state</code>
Retention Rate	<code>1 - churn_rate</code>
ARPU	<code>SUM(monthly_charges) / COUNT(active_customerid)</code>
Average Customer Tenure	<code>AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))</code>
Revenue at Risk	<code>SUM(monthly_charges) WHERE churn_score > 70</code>
Escalation Rate	<code>(SUM(escalations) / COUNT(complaints)) * 100</code>
Avg Complaints / Customer	<code>COUNT(complaints) / COUNT(DISTINCT customerid)</code>
Correlation: Escalation → Churn	<code>churn_rate WHERE escalations ≥ 1 vs 0</code>

Key Insights & Revenue Leakage

28.6%

OVERALL CHURN RATE

55.6%

MONTHLY PLAN CHURN

18.0%

% REVENUE LOSS

₹2,047

CLTV EROSION

Churn & Contract Disparity

- ✓ **Contract Gap:** Monthly contract subscribers churn at 55.6%, which is 6.7x higher than annual plan subscribers (8.3%).
- ✓ **Plan Distribution:** Bulk of churn is concentrated in the **Basic Subscription Plan**, minimizing immediate severe revenue impact.
- ✓ **Tenure & ARPU:** Average customer tenure stands at 1,451 days with an ARPU of ₹18.8.

Geo & Revenue Concentration

- ✓ **Spike Timeline:** Highest churn spike occurred during **September 2024**, with **Karnataka** being the most impacted state.
- ✓ **MRR Leakage:** Directly attributed \$73.94/mo in MRR leakage and ₹74 total revenue loss to six high-risk customers.
- ✓ **Support Correlation:** Support escalations strongly correlate with churned status vs retained cohorts.

Strategic Action Items & Next Steps



Investigate Regional Spike (Karnataka)

Conduct a deep-dive analysis on the September 2024 churn surge in Karnataka to identify price hikes, service outages, or localized technical issues.



Audit Basic Plan & Pricing Changes

Evaluate recent pricing or tier modifications on the Basic plan to determine why monthly subscribers are cancelling at a 6.7x higher rate than annual users.



Track Competitor Migration

Monitor competitive offerings and migration drivers after observing user movement to competitors due to content dissatisfaction or price sensitivity.



Target High & Medium Risk Cohorts

Prioritize high-LTV customers flagged with high/medium churn scores for proactive outreach across Email, SMS, and Calls to resolve pending complaints.